

MICHIGAN CITY ENTRANCE LIGHT, IN, USA

WMO: 997339

Lat: 41.7290N

Lon: 86.9117W

Elev: 197

StdP: 98.98

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.1	-12.5	N/A	N/A	N/A	N/A	N/A	N/A	18.6	-5.7	17.5	-4.9	7.2	190	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.2	30.8	N/A	29.4	N/A	28.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 16.5	14.7	12.7	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-1.8	-3.2	3.1	9.4	15.1	20.1	23.2	22.2	18.6	12.6	6.0	1.1
	DBStd	10.18	5.46	5.60	5.67	4.76	5.16	3.97	3.32	2.71	3.65	4.52	5.27	5.05
	HDD10.0	1488	367	370	229	69	11	0	0	0	0	21	141	280
	HDD18.3	3269	624	603	473	272	131	27	3	3	39	188	371	535
	CDD10.0	1711	1	0	16	50	168	304	410	379	258	102	21	3
	CDD18.3	449	0	0	1	3	28	81	155	123	47	11	0	0
	CDH23.3	2577	0	0	6	22	199	466	1023	584	223	53	0	0
	CDH26.7	609	0	0	0	2	54	106	278	115	46	9	0	0
(14) Wind	WSAvg	6.1	7.4	6.7	6.3	6.5	5.4	4.6	4.7	4.6	5.4	6.9	7.4	7.0
(15) Precipitation	PrecAvg	998	65	53	64	90	94	110	103	101	86	89	78	65
	PrecMax	1442	146	107	131	172	174	254	184	248	346	226	190	126
	PrecMin	759	20	20	21	27	16	38	40	22	18	21	14	21
	PrecStd	156	32	25	29	39	47	54	37	58	66	52	37	26
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.4	14.6	24.5	26.3	30.9	31.3	33.0	31.5	30.5	28.2	20.7	16.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	11.4	9.2	20.3	23.1	28.2	29.5	31.1	29.6	27.8	24.8	18.1	13.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	8.5	5.8	16.2	20.5	25.5	27.7	29.7	27.9	25.9	22.2	15.7	10.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	5.7	3.3	12.0	17.7	23.0	26.1	28.1	26.4	24.0	19.7	13.6	8.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.2	6.8	7.9	7.9	6.9	6.2	5.9	6.4	6.6	6.5	5.4
		MCDBR	9.0	9.5	12.5	12.6	11.2	9.1	8.1	8.1	8.3	9.5	9.0	8.1
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.307	0.330	0.348	0.402	0.432	0.436	0.450	0.447	0.421	0.359	0.324	0.305
	taud		2.431	2.348	2.373	2.251	2.227	2.265	2.225	2.229	2.264	2.440	2.492	2.489
	Ebn at Noon		842	872	899	868	848	842	825	815	809	830	817	816
	Edh at Noon		78	99	108	132	138	134	138	133	120	90	74	68
(44) All-Sky Solar Radiation	RadAvg		1.51	2.34	3.56	4.65	5.30	6.08	6.13	5.32	4.38	2.85	1.77	1.26
	RadStd		0.15	0.30	0.28	0.39	0.47	0.38	0.30	0.31	0.36	0.30	0.23	0.14

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page